AI Subscription &

Auto-Debit Intelligence System

SubIntel

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2nd April 2026

Problem Statement

Why does this system need to exist?

Key Customer Pain Points

- Users forget active subscriptions — money leaks silently
- Auto-debits fail due to insufficient balance
- Subscriptions buried in transaction history — hard to spot
- No advance warning before next debit date
- Failed transactions lead to penalties & merchant revenue loss

Solution Overview

What SubIntel builds:

NLP Detection

TF-IDF + Logistic Regression identifies subscription transactions with 99.81% accuracy

Pattern Engine

Detects monthly/weekly recurring patterns across 74,582 confirmed subscriptions

ML Prediction

ARIMA(1,0,0) + Linear Regression predicts next debit date & amount for 3,068 entries

Risk Scoring

Gradient Boosting classifies customers into Low / Medium / High debit failure risk

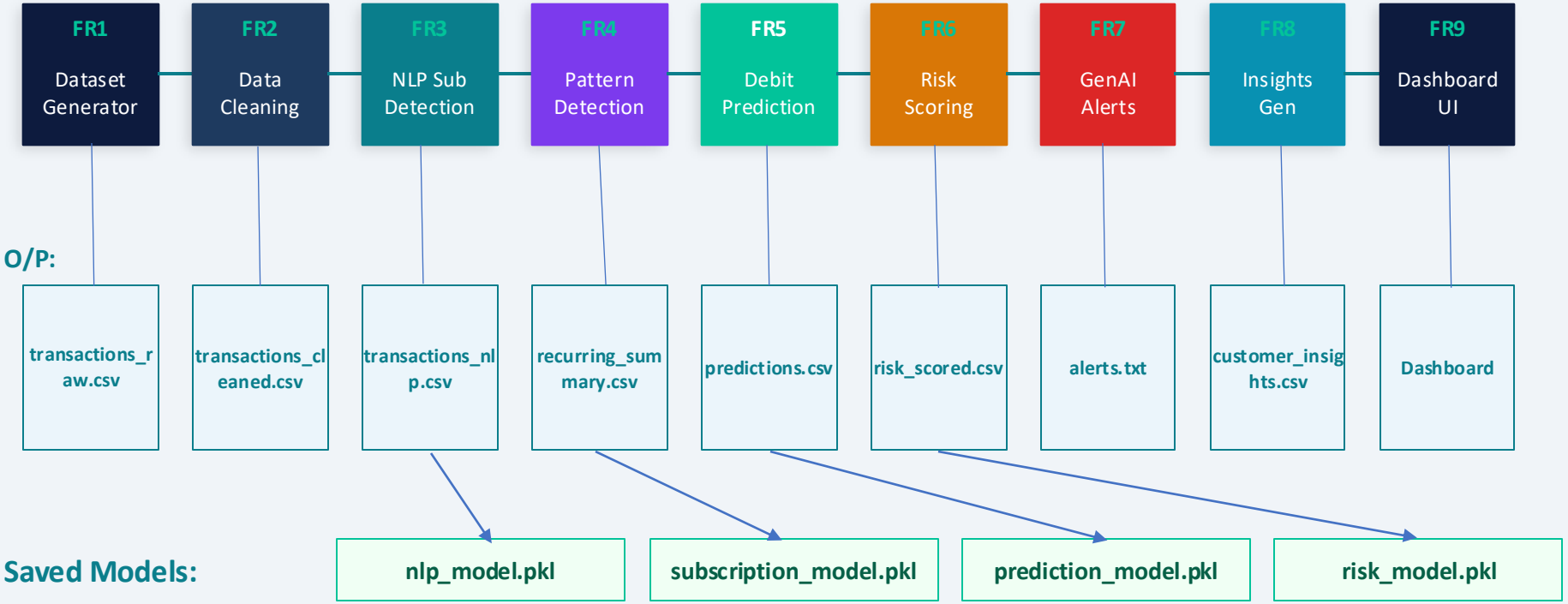
GenAI Alerts

Structured Phi-2 powered alerts with actionable suggestions — 1,200 alerts generated

Dashboard

Streamlit + Plotly interactive UI with login, filters, risk cards, and spend charts

System Architecture



Technology Stack

Why each tool was chosen

Python 3.11
Core Language

Rich ML/NLP ecosystem, fast prototyping

Faker (en_IN)
Data Generation

Realistic Indian banking transaction simulation

Pandas + NumPy
Data Processing

Fast vectorised operations on 137K rows

spaCy + TF-IDF
NLP (FR3)

Lemmatization + n-gram text features for sub detection

Scikit-learn LR
Classifier (FR3)

Fast, interpretable; 99.81% accuracy on imbalanced data

ARIMA(1,0,0)
Time Series (FR5)

AR(1) optimal for stationary subscription amounts

Gradient Boosting
Risk Model (FR6)

Handles non-linear risk features; 95th-pct normalization

Streamlit + Plotly
Dashboard (FR9)

Rapid interactive UI with real-time chart rendering

Data — Generation & Structure

Synthetic banking dataset built with Faker

Dataset Fields (FR1 + BRD §12)

CustomerID	Unique customer identifier
TransactionID	Unique transaction UUID
Date	Jan 2023 – Jun 2024 (18 months)
Description	Merchant-style transaction text
Amount	Realistic debit amounts (₹)
Balance	Running account balance
Merchant	OTT / Utility / SaaS vendor
SubscriptionFlag	Binary label (0 / 1)
Status	SUCCESS / FAILED

Dataset at a Glance

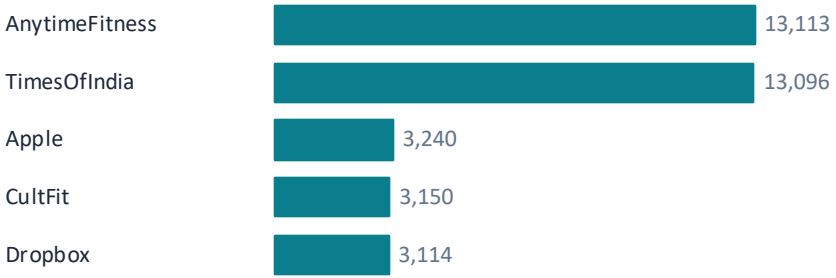
137,806
Total Transactions

1,200
Unique Customers

75,428
Subscription Rows (54.7%)

18 mo
Date Range (2023–2024)

Top Merchants by Volume



Pipeline Walkthrough

1

FR3 NLP Subscription Detection

TF-IDF (1–3 gram, 8,000 features) + Logistic Regression pipeline. spaCy lemmatization for text normalisation. False-positive filter for salary/refund keywords. Accuracy: 99.81%.

2

FR4 Recurring Pattern Detection

Groups transactions by CustomerID + Merchant. Confirms recurring if ≥3 occurrences. Detects monthly/weekly frequency. 74,582 patterns confirmed.

3

FR7+8 Alerts & Insights

Structured Phi-2 style alerts with merchant, amount, risk, and suggestion. FR8 generates plain-language insight messages per customer.

4

FR9 Streamlit Dashboard

Login page → sidebar nav → global filters → 4 sections (Active Subs, Upcoming Debits, Risk Alerts, Monthly Spend) → popup risk cards → footer.

Dashboard Demo — FR9 Streamlit UI

Login → Sidebar → Filters → Risk Cards → Spend Charts

Login Page

Username/password authentication
Sessions state via st.session_state

Active Subscriptions

Filterable table of detected subs
Merchant, frequency, amount columns

Upcoming Debits

Next debit date + predicted amount
Sorted by urgency (nearest date first)

Risk Alert Cards

Popup-style cards per customer
Colour coded: High Med Low
259 High / 595 Medium / 346 Low

Monthly Spend Chart

Plotly bar chart — spend per month
Drilldown by merchant category
Total: configurable by customer

GenAI Alerts (FR7)

Structured Phi-2 alerts panel
Merchant | Amount | Risk | Suggestion
1,200 alerts across 1,200 customers

Model & AI Results

Real metrics from the trained pipeline

FR3 — NLP Subscription Detection

99.81% Accuracy (Target: >90% ✅ Exceeded)

Model: TF-IDF (1–3 gram) + Logistic Regression
Vectorizer: 8,000 features, sublinear TF
False-positive filter: salary, refund, bonus keywords

FR5 — Debit Prediction

3,068
Predictions

99.97%
Used ARIMA

ARIMA(1,0,0) — AR(1) optimal for stationary subscription series
Date blend: 70% Linear Regression + 30% Median gap

FR6 — Risk Distribution (1,200 Customers)

FR7 — GenAI Alerts

1,200
Alerts Generated

⚠ **Upcoming Debit:** Amazon ₹499 | Risk: Medium balance
Suggestion: Add funds before 2 days

Challenges Faced

Problem: **Risk Score Collapse (FR6)**

Gradient Boosting predict_proba outputs were near-certain (all 0.0 or 1.0), causing scores to cluster at extremes

Fix: Replaced absolute thresholds with 95th-percentile dataset-relative normalization anchors

Problem: **Wrong ARIMA Order (FR5)**

Initial ARIMA(1,1,1) applied differencing to already-stationary subscription amounts → degraded forecasts

Fix: Corrected to ARIMA(1,0,0) — no integration term needed. 99.97% predictions now use this model

Problem: **NLP False Positives (FR3)**

Salary credits, NEFT/IMPS transfers, and refunds were flagged as subscriptions by the TF-IDF model

Fix: Added explicit false-positive filter set: {salary, refund, neft, imps, rtgs, dividend, bonus}

Key Learnings

NLP Pipeline Design

Building an end-to-end text classification pipeline — from raw descriptions to production-ready saved models — with spaCy lemmatization and TF-IDF vectorization.

Time Series Fundamentals

Understanding stationarity and choosing the right ARIMA order matters enormously. ARIMA(1,0,0) vs (1,1,1) changed prediction quality completely.

ML Debugging in Production

Risk score collapse taught us to always inspect model output distributions and use dataset-relative normalisation, not hard-coded thresholds.

Full-Stack ML Engineering

Connecting nine modules (data → NLP → ML → GenAI → UI) with clean file I/O contracts, saved .pkl models, and a BRD-driven architecture.

Streamlit UI Best Practices

Real-world dashboard tradeoffs: sidebar vs navbar, native containers over custom HTML, session_state for login, and keeping code under 400 lines.

Data Simulation Realism

Generating 137K synthetic transactions with realistic Indian merchant names, balance fluctuations, and failure patterns using Faker (en_IN locale).

Future Improvements

Real Bank API Integration

Connect to Plaid / RazorpayX APIs to replace synthetic data with live transaction feeds

Push Notification System

WhatsApp / SMS alerts 48 hours before predicted auto-debit using Twilio or MSG91

LLM-Powered Alert Generator

Replace rule-based Phi-2 fallback with a fine-tuned LLM for richer, contextual alerts

Subscription Cancellation CTA

Add one-click unsubscribe suggestions for redundant or low-engagement subscriptions

Multi-Bank Dashboard

Aggregate subscriptions across multiple bank accounts with de-duplication logic

Anomaly Detection Layer

Flag unexpected one-time charges masquerading as subscriptions using Isolation Forest

Thank You

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